The National Higher School of Artificial Intelligence
Introduction to Artificial Intelligence
Course Project

Release date: **14/02/2025**
Due Date: **Saturday 16/05/2024 (11:59 pm at the latest)**

Project Aim:

The aim of this project is to give you a practical experience of various techniques used for problem solving. More specifically, you are asked to use some of the (informed and uninformed) strategies based on the general Graph Search algorithm you have implemented in the lab along with a version of Constraint Satisfaction Problem (CSP) as well as other problem solving techniques. These will be used to solve one of the real-life problems suggested below.

You are also asked to provide specific visualisation depending on the problem you are assigned to tackle.

Your work should be returned typed as a pdf file and submitted on Google Classroom under your group as explained below.

General Notes:

You are required to submit a complete report which, for each aspect of the project, gives the needed explanations and results of the execution of your implementation. A good report is not one which explains the code; it rather explains the solutions and any decisions taken to make the design workable and effective. It will also present the problem that is considered and should also give convincing comparatives of executions using different techniques. The well-commented codes will be submitted under electronic format in one package according to the directives given below.

The project is to be worked on by **teams of 6 students**, one of whom will be designated as the Team Leader and is the one who will be in charge of any communication with the course instructors. In your report, you will include a table that will indicate precisely who did what on the project. **Bear in mind that it is not acceptable that a student work only on report writing; all the students will have to get involved in all aspects of the project.** (We will be strict on this!) Each team will give a demo and the team members will not necessarily have the same grade; it depends on their individual contributions and their answers to questions during the demo as detailed below. The demos will be given at ENSIA in shaa Allaah.

Practical Aspects:

As stated above, the work must be done in teams of 6 students. Any other grouping will not be allowed and will be penalised unless exceptionally allowed in writing by email (say because not enough students are left to form a team of 6).

You are asked to do the following:

1. Your team leader will fill in the Google Form available at https://docs.google.com/forms/d/e/1FAIpQLScnsmb0ucrfSD-2Z4BFMbQzYZfHV7IM_sAwpGo21TAMVQAbow/viewform?usp=publish-editor

   Deadline: Thursday 19/02/2026 at 11:59 p.m.
   The form allows the leader to provide the requested details of the members of your team. **If you fail to have an acceptable number of members (i.e. EXACTLY 6) in your team, your project grade will be heavily penalised.**

2. Any future communication will be with team leaders ONLY (unless one of the instructors believes he/she needs to include the rest of the team, for whatever reason).

3. You will be informed by **Saturday 21/02/2024**, in shaa Allaah, of the mini-project topic assignments for the various 6-member teams.

4. Mini-project work submission: The report, the dataset used (if it is collected by you, not taken from the internet) and the code will be submitted onto Google Classroom as ONE COMPRESSED FOLDER whose name **is the team leader's name and the project number**. **It is your responsibility to ensure that no files are missing and are submitted within the deadline.** Failure to do so, your project will be penalised. **If your submission is empty or files are missing, you will get 0 for the relevant parts of the project mark.**

Deliverables:
As mentioned above, you will submit on Google Classroom ONE COMPRESSED FILE which contains your report, and ONE Jupyter Notebook which will contain the code that implements all the search techniques, CSP, GA, etc., for problem solving, AND the dataset used if you have collected it locally (see the problem descriptions). You do not have to submit the web/mobile application files (if any); you will get a chance to show your app the day of the demo.

1. The report must be such that you
   a) Clearly explain the problem and the solutions you provide in your project. (See the attached "Intro_to_AI_project report template_s2025-2026.docx".)
   b) Explain in full detail your design choices for your solution, and the results, including summary/comparative tables, etc.
   c) At the end of the report, explain in a table who did what very precisely (points will be deducted if the distribution of responsibilities is not specified in the report or if the answer is "all members worked together on all the project parts".) The tasks may be related to writing the report (sections), data collection and curation, solution design, implementation, front/back end programming, testing, etc. **Each team member MUST be involved in all aspects of the project, i.e. design, implementation, and writing up**.

Observe all the rules mentioned in this document.

2. Your Python Jupyter Notebook will be uploaded onto Google Classroom as part of the ONE compressed file as explained above. In all cases:
   - The Jupyter Notebook should have text cells that clearly explain various aspects of the code and some brief explanation about the data you have used, which must be placed in a separate folder named ***Data***. The data files must be meaningful.
   - Do not expect your instructor to correct or debug your code. It must run right away your data being placed in the same folder as the Jupiter notebook.
   - The names of any files and folders must be as intuitive and meaningful as possible.
   - Respect the good software development practices you have been taught for all your code.

Project submission date and time: Saturday 16/05/2024 no later than 11:59 pm. (Recall that submissions are time-stamped to the second on Google Meet.)
- For every 24 hours of late submission from the deadline, 25% of the obtained grade will be deducted.
- No report will be accepted after Wednesday 20/05/2025 11:59 p.m.
- Note that even after having submitted your project report and code, you are entitled to continue improving your code/app till the demo day (hour!).

Important Note:
A good report does not have to be long! It is up to you to decide how to have a sufficiently complete report that remains as short as possible (so that you do not get penalised for its length). In other words, it is not a question of filling in but of making a written presentation as scientific and clear as possible. **To help you, I limit the length of the report to 13 pages with font Times New Roman size 11. Tables of results and/or comparisons must be included in the report and discussed.** The Appendix can only contain

screen shots about the application or search results. The pages of the Appendix do not count as part of the 13 pages.

Any pages beyond 13 (the Appendix excluded), will be penalised: 0.5/20 which will deducted for each extra page.

The final project grade for your team will be relative to that of the other teams who will be assigned the same topic as you!! So you are competing with those teams.

Grading of the mini-project :

Mini-project part	Grading Scheme	Points (Total 20)
Report including analysis and discussion of the results	25%	/5
Formulation of the problem and DATA (in reasonable quantity and quality) folder and ease of integration of these formulations in the package. (Note that it is a General Search Algorithm in a large part of this project, hence one should easily be able to apply your implementation to other problems)	15 %	/3
Overall solution (the package functionalities, richness, friendliness, visualisation of search, etc.) and code quality	30 %	/6
Demo and Questions&Answers	30 %	/6

Final notes:

- **Students in the same team will not necessarily have the same grades since at least the quality of their answers to demo questions will vary.**
- **It is your RESPONSIBILITY to ensure that your team leader has uploaded your package onto Google Classroom. No excuse will be accepted for any failure to do so, in which case the corresponding marks will be zeroed.**
- **Any submissions from a member who is not a team leader will be penalised.**
- **No email submission will be accepted.**
- Respect ALL the above directives.

Enjoy! ☺